

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Ms. Mohammed Razia Alangir Banu
Student Name	Katta Yamini Sai
Roll Number	2520030380
Branch	CSE
Course Outcome	CO-1

Case Study Topic: Pet Record Indexing using AVL Tree

Work Done Summary:

In this case study, an AVL Tree was implemented for managing pet health records in the PetCare Smart Veterinary Clinic & Pet Health Management System. The AVL Tree stores pet IDs and maintains balance automatically after every insertion and deletion operation. Various balancing operations such as LL, RR, LR, and RL rotations were performed whenever the balance factor exceeded the allowed limit. The implementation demonstrates efficient searching, insertion, and deletion with $O(\log n)$ time complexity.

Implementation Details:

1. Created an AVL Tree node structure to store pet record information.
2. Inserted pet records into the AVL Tree.
3. Calculated height and balance factor for every node.
4. Applied LL, RR, LR, and RL rotations when required.
5. Implemented search operation for pet record lookup
6. Implemented deletion operation while maintaining tree balance.
7. Compared AVL Tree performance with a normal Binary Search Tree.
8. Verified that AVL Tree maintains $O(\log n)$ search performance.

RESULT : SCREENSHOTS/OUTPUT

```

Sample Output

Pet Records:
Pet ID: 101 | Pet Name: Tommy
Pet ID: 102 | Pet Name: Lucy
Pet ID: 103 | Pet Name: Rocky
Pet ID: 104 | Pet Name: Max
Pet ID: 105 | Pet Name: Bella

Pet Found: Rocky

After Deletion:
Pet ID: 101 | Pet Name: Tommy
Pet ID: 103 | Pet Name: Rocky
Pet ID: 104 | Pet Name: Max
Pet ID: 105 | Pet Name: Bella

```

This case study implements an AVL Tree for storing and managing tourist destinations in the TourPlan system. The AVL Tree maintains balance through rotations, ensuring efficient insertion, deletion, and search operations with $O(\log n)$ time complexity. The results demonstrate that AVL Trees provide faster and more reliable performance than normal BSTs for large tourism datasets.

GitHub link:https://github.com/yaminikatta18/DSA_CASE-STUDIES/commit/667ac329e98ab48ee3d5a3cf05cd309d38fe5fe8

Student Signature	Faculty Signature



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Case Study Topic: Pet Health Range Query System using Segment Tree Done

Summary:

1. In this case study, a Segment Tree was implemented for managing pet health records in the PetCare – Smart Veterinary Clinic & Pet Health Management System. The Segment Tree efficiently processes range minimum queries to help veterinarians identify pets with the lowest health scores within a specified range. The implementation supports fast query processing and dynamic updates, making it suitable for veterinary clinics where pet health data changes frequently.
2. Implementation Details:
 1. Created an array to store pet health scores.
 2. Constructed a Segment Tree from the pet health data.
 3. Implemented range minimum query operations
 4. Allowed users to search pet health scores within a specified range.
 5. Supported health record updates without rebuilding the entire structure.
 6. Compared Segment Tree performance with linear search.
 7. Verified query execution efficiency.
 8. Demonstrated $O(\log n)$ query and update operations.

RESULT : SCREENSHOTS/OUTPUT

```
Output
=====
PETCARE HEALTH RANGE QUERY SYSTEM
=====

Pet Health Scores:
85 72 90 60 78 95 68 88

Range Query:
Minimum Health Score from index 2 to 6 = 60

Updating Health Score...
Pet at index 3 health score changed from 60 to 55

After Update:
Minimum Health Score from index 2 to 6 = 55

=====
Time Complexity:
Segment Tree Construction :  $O(n)$ 
Range Query                :  $O(\log n)$ 
Update Operation           :  $O(\log n)$ 
=====
```

This case study implements a Segment Tree for efficient pet health range queries in the PetCare system. The Segment Tree enables fast range minimum searches and supports dynamic updates with $O(\log n)$ complexity. The results demonstrate that Segment Trees are highly effective for veterinary applications involving interval-based health analysis. Time Complexity: Segment Tree Construction : $O(n)$ Range Query : $O(\log n)$ Update Operation : $O(\log n)$

Github Link:https://github.com/yaminikatta18/DSA_CASE-STUDIES

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Course Outcome	CO-3

Case Study Topic: Case Study Topic: PetCare – Smart Veterinary Clinic & Pet Health Management System using BFS, DFS and Prim's Minimum Spanning Tree

Work Done Summary:

A graph-based veterinary clinic network was developed for the PetCare System. Veterinary departments, pet service units, and clinic branches were represented as graph vertices, while communication and service connections were represented as weighted edges. BFS and DFS algorithms were implemented to explore connectivity among clinic services, and Prim's Algorithm was used to construct a Minimum Spanning Tree (MST) to identify the minimum-cost network connecting all departments. The implementation demonstrates graph traversal and network optimization techniques used in modern pet healthcare management systems.

Implementation Details:

1. Created a graph representation of veterinary departments and clinic branches using an adjacency matrix.
2. Added weighted edges to represent communication or service costs between departments.
3. Implemented Breadth First Search (BFS) to explore connected services level by level.
4. Implemented Depth First Search (DFS) to traverse all reachable departments recursively.
5. Applied Prim's Algorithm to construct a Minimum Spanning Tree (MST).
6. Calculated the minimum network cost connecting all veterinary services.
7. Displayed BFS traversal, DFS traversal, and MST results.
8. Verified connectivity and optimized service coordination across the PetCare system.

RESULT : SCREENSHOTS/OUTPUT

```
Sample Output

PetCare Smart Veterinary Clinic System
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BFS Traversal: 0 1 2 3 4

DFS Traversal: 0 1 2 3 4

Minimum Spanning Tree (Prim's Algorithm):
Edge          Weight
Vaccination - Consultation    1
Registration - Vaccination    2
Consultation - Pharmacy       5
Pharmacy - Emergency          2

Total Minimum Cost = 10
```

This case study implements BFS, DFS, and Prim's Algorithm to analyze connectivity within the PetCare system. Graph traversal techniques efficiently explore veterinary services and clinic branches, while Prim's Algorithm identifies the minimum-cost network connecting all departments. The results demonstrate the importance of graph algorithms in healthcare management, resource allocation, and service optimization

Github Link:

https://github.com/yaminikatta18/DSA_CASE-STUDIES/blob/main/CO3.java

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